

Article

Control Strategy of Multiple Battery Energy Storage Stations for Power Grid Peak Shaving

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Abstract

In order to achieve the goals of carbon neutrality, large-scale storage of renewable energy sources has been integrated into the power grid. Under these circumstances, the power grid faces the challenge of peak shaving. Therefore, this paper proposes a coordinated variable-power control strategy for multiple battery energy storage stations (BESSs), improving the performance of peak shaving. Firstly, the strategy involves constructing an optimization model incorporating load forecasting, capacity constraints, and security indices to design a coordination mechanism tracking the target load band with the equivalent power. Secondly, it establishes a quantitative evaluation system using metrics such as peak–valley difference and load standard deviation. Comparison based on typical daily cases shows that, compared with the constant power strategy, the coordinated variable-power control strategy has a more obvious and comprehensive improvement in overall peak-shaving effects. Furthermore, it employs a “dynamic dispatch of multiple BESS” mode, effectively mitigating the risks and flexibility issues associated with single BESSs. This strategy provides a reliable new approach for large-scale energy storage to participate in high-precision peaking.

Keywords: battery energy storage stations; peak shaving; power grid; variable power; control strategy



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1. Introduction

Renewable energy has emerged as a critical pathway for achieving the goals of carbon neutrality [1]. The intermittent and stochastic nature of renewable energy, combined with its rising grid share, has increased peak-shaving pressure on power systems [2]. As for an urban power grid, the predominance of industrial loads has resulted in a progressive widening of the diurnal peak-to-valley load differential [3]. This context emphasizes the fundamental importance of peak-shaving and valley-filling strategies as core mechanisms within load management for optimizing overall system operation. The conventional approaches employed, such as capacity expansion of generation units and transmission infrastructure, achieve their objectives at the expense of grid operational efficiency, particularly concerning economic efficiency and resource utilization. Consequently, traditional peak-shaving measures are proving increasingly inadequate to effectively mitigate the exacerbating load peak–valley disparity and the concomitant decline in power supply reliability. To alleviate the mounting pressure on the power system supply–demand equilibrium, the

implementation of more sophisticated, efficient, and reliable demand-side management strategies is imperative.

Under the circumstance, battery energy storage stations (BESSs) offer a new solution to peak regulation pressure by leveraging their flexible “low storage and high generation” capabilities and rapid response [4]. Compared to traditional thermal power peak load regulation, BESSs feature millisecond-level response speeds. By leveraging their bidirectional energy flow characteristics, they can regulate load peak-to-valley differences, thereby smoothing out load fluctuations [5]. From a technical and economic perspective, the cost of lithium-ion battery energy storage has decreased by over 60% in the past five years, making it more widely applicable in daily peak-shaving scenarios [6]. More importantly, BESSs can be deployed in a distributed manner at load centers, directly alleviating transmission and power-grid congestion and reducing network losses, which further demonstrate the advantages [7].

Previous studies on BESS participation in peak–valley regulation have been conducted, primarily focusing on two different strategies: one involves BESSs directly participating in grid peak–valley regulation, whereas the other involves BESSs indirectly participating in grid peak–valley regulation [8]. In the first strategy, the energy storage station directly participates in peak–valley regulation, with peak–valley regulation and valley filling as the core optimization objectives [9], aiming to establish an optimal charging and discharging strategy. Such studies directly address the effectiveness of peak regulation and typically incorporate actual operational constraints. For example, models may consider optimizing peak–valley regulation while reducing grid losses [10] or accounting for limitations on charge–discharge cycle depth [11]. In complex systems, models must also integrate the peak–valley periods of the main grid and system power balance to formulate energy storage scheduling strategies [12]. Some studies employ simplified constant-power charging and discharging models to achieve peak–valley regulation objectives or conduct economic evaluations [13]. The algorithms for addressing such direct peak-shaving strategies are more diverse. Due to their ability to handle complex nonlinear problems [14], intelligent algorithms such as genetic algorithms and particle swarm algorithms are widely used [15]. In addition, active control strategies combining short-term load forecasting have been proposed for grid energy storage optimization [16].

The second strategy involves BESSs indirectly participating in peak–valley regulation. Although peak–valley regulation is not directly set as an optimization objective, it can be indirectly achieved by constructing a refined economic model to guide the charging and discharging behavior of the energy storage station. For example, by utilizing time-of-use pricing mechanisms, the energy storage station can charge during off-peak hours and discharge during peak hours to maximize economic benefits [17], thereby indirectly achieving peak–valley regulation. Related economic models include the BESS revenue model [18], which considers the interests of the generation side, grid side, and user side [19], and the investment economic model [20], which analyzes the economic conditions for large-scale energy storage applications from an investment return perspective.

In summary, although existing strategies for BESSs to participate in peak shaving can smooth load fluctuations, most of them are still limited to a single BESS, and their effectiveness is restricted. Single BESSs are limited by their rated power and capacity constraints, making it difficult for their regulation capabilities to match continuously changing load curves. To overcome this inherent limitation and improve peak-shaving effectiveness, this paper proposes a multiple BESS coordinated variable-power control strategy. The main contributions of this paper are as follows:

This paper proposes a control strategy of multiple battery energy storage stations (BESSs) for power-grid peak shaving. Firstly, the working principle of the variable-power

control strategy for multiple BESSs is proposed. Secondly, an optimization model is constructed incorporating load forecasting, safety constraints, and evaluations for peak-shaving effectiveness. Thirdly, the collaborative algorithm based on a continuous time-period search and phased control is used to obtain the optimal control strategy. Finally, a typical daily case is employed to quantitatively analyze the improvement effects on peak shaving.

The structure of this paper is as follows: Section 2 explains the design principles of the cooperative variable-power mechanism and constructs an optimization model incorporating load forecasting, safety constraints, and evaluation metrics. Section 3 conducts a detailed analysis of the peak-shaving effects of the cooperative variable-power strategy and the single constant power strategy through a comparison of typical daily cases, quantifying the improvement effects on key metrics. Section 4 summarizes the work of this paper and outlines future research directions.

2. Methodology

2.1. Peak Load Shifting Control Strategies

There are two operating modes for BESS in power systems. During low load periods, the BESS switches to a charging state and acts as an electrical load, absorbing excess energy from the load absorption system. During peak load periods, the BESS switches to the discharging state and acts as a generator, discharging excess energy into the system. Developing a reasonable BESS charging and discharging control strategy can help to improve the peak-shaving efficiency and BESS utilization.

2.1.1. Control Strategy for Constant Power Charging and Discharging Mode

The constant power charging and discharging control strategy operates on a daily time scale and minimizes load fluctuations. Assuming that the charging and discharging power of the BESS remains constant, the start and stop times for charging/discharging are controlled by real-time load values, with the specific operational mode illustrated in Figure 1. After regulation by the BESS's charging and discharging processes, load fluctuations will be maintained between the peak-shaving line and the valley-filling line, thereby achieving peak-shaving and valley-filling functionality.

Under this strategy, since the load power is typically much larger than the energy storage power, the BESS charges and discharges at its rated power. The charging and discharging times of the BESS can be calculated based on the total capacity and rated charging/discharging power. Additionally, the BESS charges and discharges uniformly throughout the day, and the number of charging and discharging cycles can be determined based on the load curve.

The specific steps are as follows: first, based on the charging and discharging power P_{charge} and $P_{discharge}$ and the capacity E of the BESS, the charging and discharging times T_{charge} and $T_{discharge}$ are determined. After determining the charging and discharging times, draw a line L parallel to the horizontal axis based on the lowest point and the peak point of the known load curve. Set the step size for moving along the positive direction of the axis to ΔL and calculate the distance d between the two intersection points of line L and the curve. When $d = T_{charge}$, the time interval between these two intersection points is the desired reasonable charging time range. If not, continue moving line L along the positive direction of the Y-axis until the reasonable charging time range is determined. Similarly, draw a horizontal line at the minimum load point and gradually move it upward in the same step size L until the total time interval between the horizontal line and the load curve equals T , thereby determining the discharge time range. The flowchart of the constant power control strategy is shown in Figure 2.

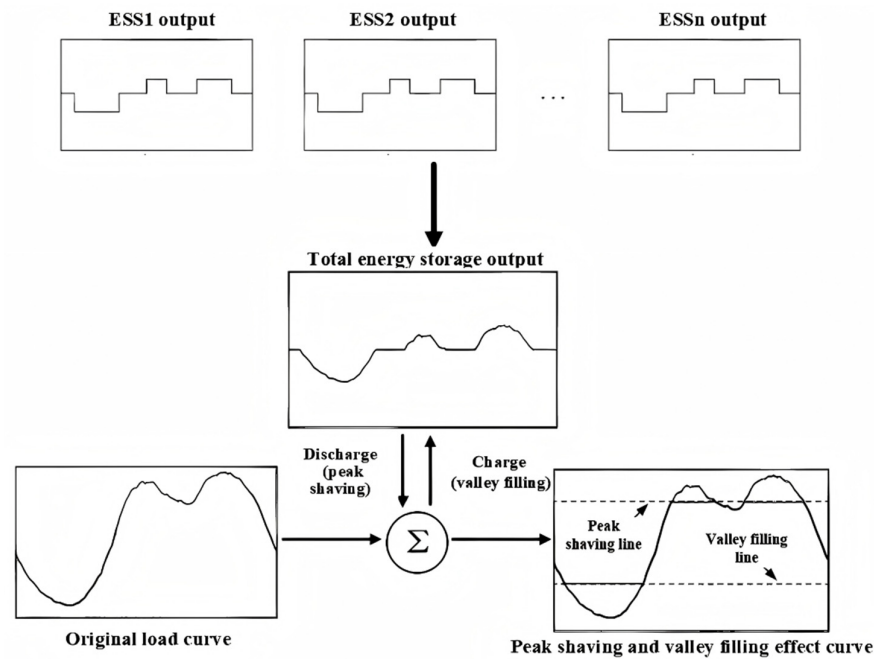


Figure 1. Constant power control strategy for BESS participating in peak shaving in power grids.

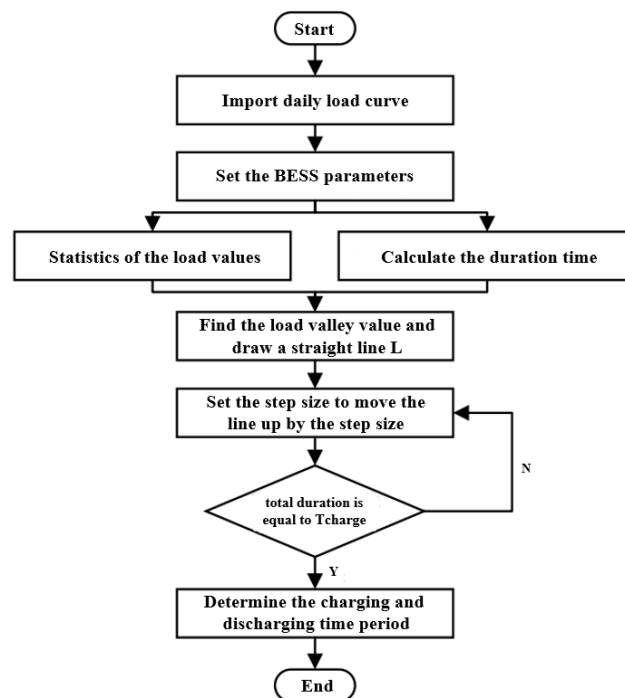


Figure 2. Constant power control strategy flowchart.

2.1.2. Control Strategy for Variable-Power Charging and Discharging Mode

The aforementioned strategy primarily targets single-storage operation scenarios. However, when implementing peak shaving and valley filling at the power-grid level, the capacity, power, and flexibility of a single storage power station often fail to meet actual peak regulation requirements. Additionally, adopting a single variable-power strategy imposes high demands on equipment regulation performance. Therefore, a variable-power charge–discharge control strategy based on the coordinated operation of multiple BESSs is adopted for power grids.

The core idea of this strategy is to utilize multiple BESS units configured with constant power charging and discharging control strategies. By coordinating the start/stop

timing and charging/discharging states of each BESS unit, the strategy achieves an equivalent, dynamically varying charging and discharging power across the entire grid, thereby adapting to the continuous fluctuations in load. The objective is to constrain the strategy based on the power-grid load forecast curve and overall energy balance within a daily timeframe. By optimizing the cluster behavior of multiple constant-power BESS units, the strategy precisely tracks the peak-shaving and valley-filling curves, ultimately ensuring that the regulated load curve operates as smoothly as possible within the target range. This significantly enhances the depth, accuracy, and system flexibility of peak-shaving and valley-filling operations while reducing the dynamic regulation requirements on individual energy storage device inverters. The specific operational mode is illustrated in Figure 3.

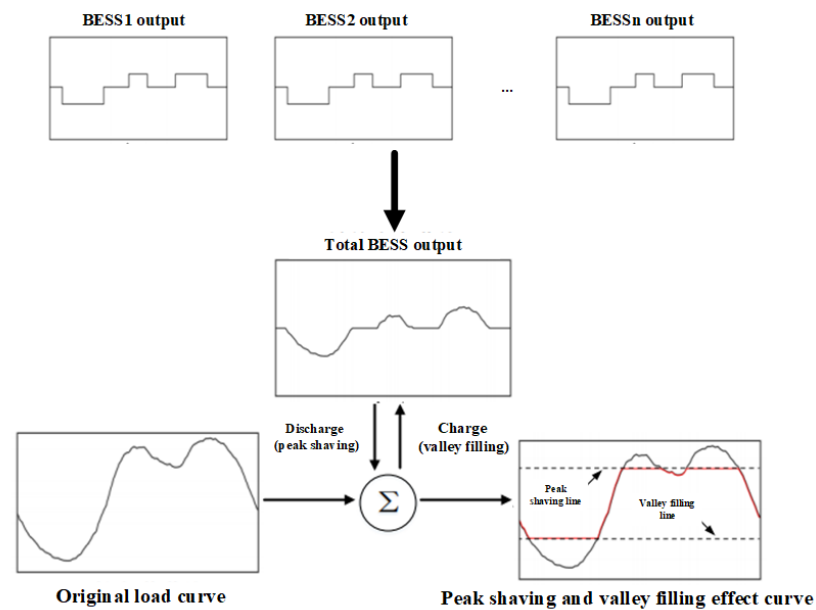


Figure 3. Variable-power control strategy for multiple BESS participating in peak shaving in power grids.

The implementation of this strategy relies on centralized coordination and control of multiple BESS units within the region. Each BESS operates independently with its own constant-power charging and discharging strategy, meaning that its charging/discharging power is fixed at its rated power, and its actions are triggered by the dispatch instructions it receives. Based on the total load forecast curve of the power grid and the overall peak-shaving and valley-filling objectives, commands are directly issued to dynamically adjust the output of each BESS participating in the cluster. Through this combination of operations of discrete constant-power BESS, a continuously varying charging/discharging power is effectively generated on the power-grid bus.

The specific steps are formalized in Algorithm 1 (Section 2.4). firstly, obtain the load forecast curve of the power grid and determine the overall peak-shaving and valley-filling targets. Secondly, obtain the parameters of each BESS participating in the cluster (rated power, available capacity, initial SOC, etc.). Thirdly, compare the forecast load curve with the target band and calculate the total charging/discharging power demand of the power-grid bus required to maintain the load within the target band for each scheduling period. Then, based on the total demand, the current status of each BESS, and the constraints, determine the operation plan for each BESS. Finally, each BESS independently executes its constant-power charging/discharging operation based on the received instructions. Figure 4 shows the flowchart of the variable-power control strategy.

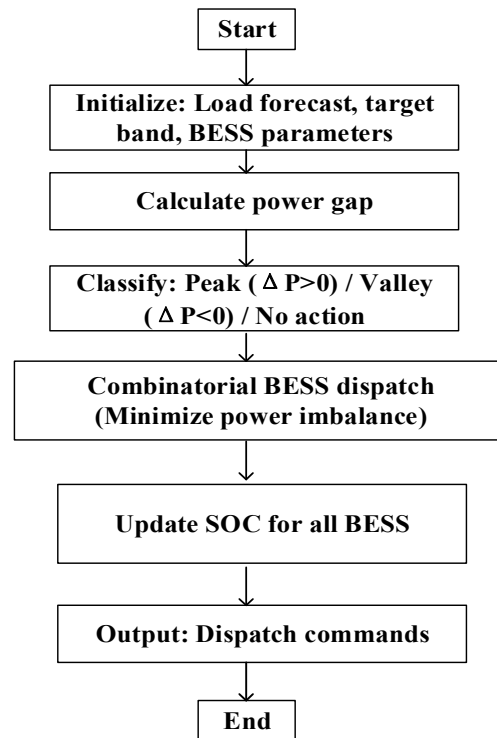


Figure 4. Variable-power control strategy flowchart.

2.2. Evaluation

In order to illustrate the effectiveness of BESS in peak shaving and valley filling and to evaluate the above control strategies, indicators for evaluating the effectiveness of peak shaving and valley filling are constructed below.

(1) Absolute peak-to-valley difference ΔP : The absolute peak-to-valley difference refers to the difference between the peak load and the valley load, which is used to characterize the maximum absolute deviation of load output on a daily time scale. The definition is as follows:

$$\Delta P = P_{\max} - P_{\min} \quad (1)$$

(2) Peak-shaving depth: The amount of power reduced or added by the BESS during peak shaving. The definition is as follows:

$$\Delta P_{\text{peak}} = P_{\text{peak}} - P'_{\text{peak}} \quad (2)$$

$$\Delta P_{\text{valley}} = P'_{\text{valley}} - P_{\text{valley}} \quad (3)$$

(3) Peak–valley coefficient α : The peak–valley coefficient is the ratio of the valley value to the peak value of the load, which characterizes the flatness of the load curve. The flatter the curve, the larger the coefficient. It is defined as follows:

$$\alpha = \frac{P_{\min}}{P_{\max}} \times 100\% \quad (4)$$

(4) Peak–valley difference rate β : The peak–valley difference rate is the ratio of the absolute peak-to-valley difference to the peak load, which indicates the range of load fluctuations. The smaller the peak–valley difference rate, the smaller the range of load fluctuations. The definition is as follows:

$$\beta = \frac{\Delta P}{P_{\max}} \times 100\% \quad (5)$$

(5) Standard deviation of load variation X : The standard deviation of load variation indicates the dispersion of load data. The smaller the standard deviation, the more concentrated the data, meaning smaller load fluctuations. The definition is as follows:

$$X = \sqrt{\frac{1}{n} \sum_{i=1}^n (P_i - \bar{P})^2} \quad (6)$$

2.3. Constraint Conditions

(1) BESS capacity constraints:

$$SOC_{\min} \leq SOC(t) \leq SOC_{\max} \quad (7)$$

where SOC is state of charge, which indicates the ratio of the battery's current remaining usable capacity to its maximum usable capacity when fully charged. This constraint ensures that the energy stored $SOC(t)$ in the BESS at any given time t must be between its minimum allowable capacity $SOC_{\min}$ (usually greater than 0 to prevent deep discharge damage to the battery) and maximum allowable capacity $SOC_{\max}$ (to prevent overcharging).

(2) Charging and discharging power limits:

$$\begin{cases} P_{charge}^{\max} \leq P_{charge}(t) \leq 0 \\ 0 \leq P_{discharge}(t) \leq P_{discharge}^{\max} \end{cases} \quad (8)$$

this constraint limits the charging and discharging power $P(t)$ of the BESS at time t . The charging power is negative, and the discharging power is positive. $P(t)$ must not exceed the rated power capacity of the equipment to prevent damage to energy storage equipment due to over current and ensure that power conversion equipment operates within safe limits.

(3) Energy conservation constraints:

$$SOC(t+1) = SOC(t) + \left(\eta_{charge} \cdot P_{charge}(t) - \frac{P_{discharge}(t)}{\eta_{discharge}} \right) \cdot \frac{\Delta t}{E_N} \quad (9)$$

this constraint ensures that the model accurately reflects the actual energy flow and losses during the charging and discharging processes of the BESS, forming the basis for precise scheduling and state estimation.

Under the constant power strategy, the charging and discharging power of the BESS remains constant. The rated power of the energy storage device should be less than the difference between the peak-shaving and valley-filling reference lines to avoid repeated judgments. At the same time, the rated power must be greater than the maximum value of the load peak-to-valley difference before and after the energy storage action to prevent the BESS from acting multiple times. When calculating the energy storage capacity, a certain margin should be retained for the battery state.

In the variable-power strategy, the charging and discharging power of the energy storage device adjusts according to load changes, so there is no need to consider the scenario where the load curve exceeds the peak-shaving and valley-filling baseline after the energy storage device operates. Additionally, since the power is variable, the battery capacity should be calculated based on the actual power.

2.4. Algorithm Implementation

The proposed Algorithm 1 (Multi-BESS Coordinated Variable-Power Control) is implemented as follows:

Algorithm 1 Multi-BESS Coordinated Variable-Power Control for Grid Peak Shaving

(1) Initialization

Input: $L_{\text{forecast}}(t)$: Forecasted grid load curve [$t = 1:T$]; Target_Band: Desired load band [$L_{\text{min, target}}, L_{\text{max, target}}$]; $\text{BESS}_{\text{list}} = \{\text{BESS}_i\}$: List of BESS units (rated power, capacity, $\text{SOC}_{\text{min/max}}$)

(2) Time-Step Processing($t = 1 \sim T$)

Calculate power gap:

$$\Delta P_{\text{gap}}(t) = L_{\text{forecast}}(t) - \text{Target_Band}(t)$$

(3) Power Allocation:

If $\Delta P_{\text{gap}}(t) > 0$ (peak):

$$P_{\text{discharge, total}}(t) = \min(\Delta P_{\text{gap}}(t), \sum \text{BESS}_i, \text{discharge capacity})$$

Dispatch BESS units via combinatorial optimization to meet $P_{\text{discharge, total}}(t)$ while respecting SOC constraints (7), Power limits (8), and Energy conservation (9).

If $\Delta P_{\text{gap}}(t) < 0$ (valley):

$$P_{\text{charge, total}}(t) = \min(|\Delta P_{\text{gap}}(t)|, \sum \text{BESS}_i, \text{charge capacity})$$

Optimize BESS charging sequence (similar to Step 2).

(4) Dispatch and Update:

Send power to each BESS unit and update SOC for all BESS units by using (9).

(5) Termination:

Output: Time-series dispatch commands for all BESS units.

3. Results

Taking the daily grid load as the research object, typical daily data with a sampling interval of 15 minutes was selected. The original load data is shown in Figure 5, and the corresponding indicators are shown in Table 1. The initial configuration of the BESS is as follows: $\text{SOC} = 0.8$, $P_{\text{rated}} = 10$ MW, $T = 24$ h, and $C = 20$ MW·h. These parameter values are representative of typical utility-scale BESS installations configured for daily peak-shaving duty cycles. To compare the two control strategies, the basic configuration of the BESS remains the same under different strategies.

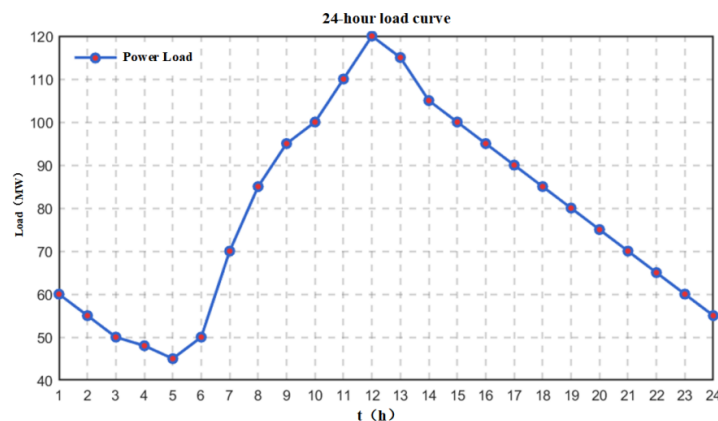


Figure 5. Typical daily raw load curve.

Table 1. Initial configuration of BESS.

Rated Capacity	Rated Power	Initial SOC	Max Charge Capacity	Max Discharge Capacity
20.0 MW·h	10.0 MW	0.80	2.0 MW·h	12.0 MW·h

3.1. Performance of Constant Power Charging and Discharging Control Strategy

The BESS charges and discharges at a constant power of 10 MW. After analyzing the data, the load peak-shaving and valley-filling variation curve under the constant power strategy can be obtained, as shown in Figure 6. The output of the BESS is shown in Figure 7, and the corresponding indicator changes are shown in Table 2.

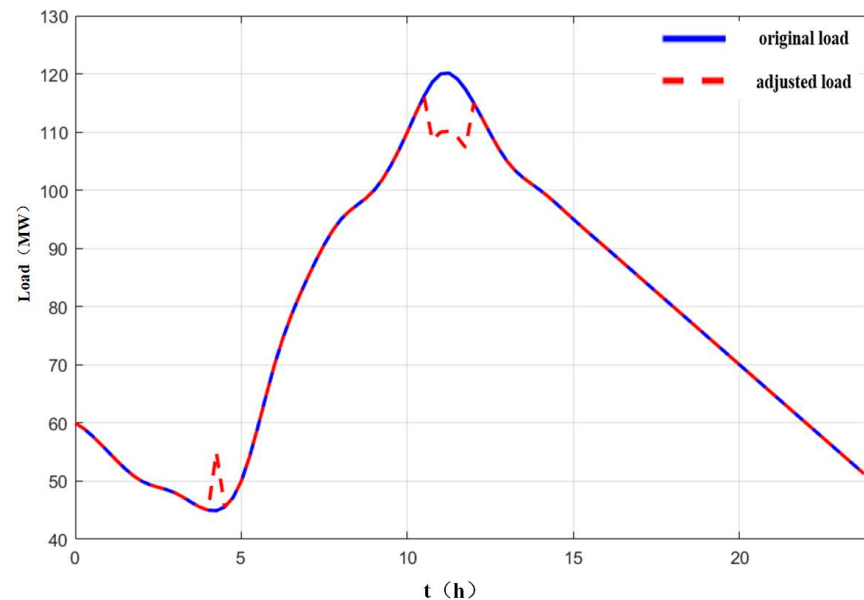


Figure 6. Changes in load curves under constant power strategy.

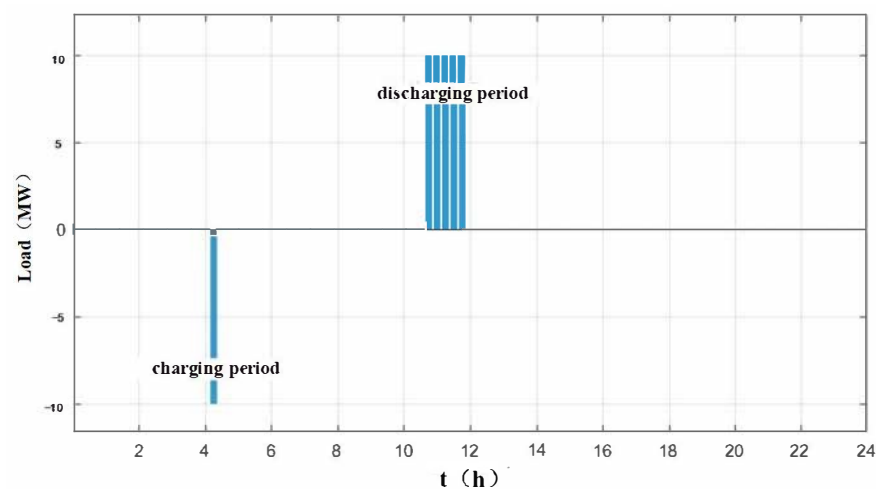


Figure 7. Schematic diagram of BESS output under constant power strategy.

Table 2. Summary of energy storage peak-shaving plans.

Time Period Type	Start Time (h)	End Time (h)	Duration (h)	Power (MW)
Charge	4.25	4.75	0.50	−10.0
Discharge	10.75	11.75	1.25	10.0

Figure 6 provides a visual comparison of the effectiveness of the BESS in peak shaving for power grids before and after the adoption of a constant power charge–discharge control strategy. The figure shows the changes in the original load curve and the adjusted load curve. The original load curve exhibits the significant fluctuations characteristic of the natural load on a typical day. After being regulated by the BESS's constant power charge–discharge control, the load curve undergoes changes: during peak periods, the

BESS discharges at a fixed power, effectively reducing peak load; during off-peak periods, the BESS charges, thereby increasing the off-peak load. Figure 6 demonstrates the effectiveness of the constant-power control strategy in achieving peak-shaving and valley-filling functionality.

Figure 7 shows the operating status of the BESS under a constant power strategy, corresponding to the scheduling plan in Table 2: two actions throughout the day and the BESS charging and discharging at rated power but exposing certain defects: first, the separation of charging and discharging periods leads to relatively low equipment utilization; in addition, the output curve reflects the poor flexibility of a single fixed-power strategy.

The constant power strategy performs charging and discharging operations at a fixed-power level during preset time periods, which achieves basic peak-shaving and valley-filling functions but has significant limitations. The peak shaving effect of BESS using a constant power strategy is shown in Table 3. Its adjustment strategy is crude, equipment utilization is low, and the strategy relies on a single-energy storage device, which is relatively inflexible when dealing with continuous load fluctuations.

Table 3. Summary of energy storage peak-shaving effects under constant power strategy.

Indicator	Before Peak Shaving	After Peak Shaving	Change
Maximum load (MW)	120.13	116.18	3.95
Minimum load (MW)	44.92	45.00	0.08
ΔP (MW)	75.21	71.18	4.03
α	0.3739	0.3873	0.0134
β	0.6261	0.6127	0.0134
X	22.8483	21.8695	0.9788

3.2. Performance of Variable-Power Charging and Discharging Control Strategy

The BESS operates at a constant power of 10 MW during charging and discharging. After analyzing the data, the load peak-shaving and valley-filling variation curves under the variable-power strategy can be obtained, as shown in Figure 8. The power output of the BESS is shown in Figure 9, and the corresponding indicator changes are shown in Table 4. The final plan for energy storage participation in peak shaving is shown in Table 5. The peak and valley values represent the peak and valley values of the load curve under the variable-power strategy.

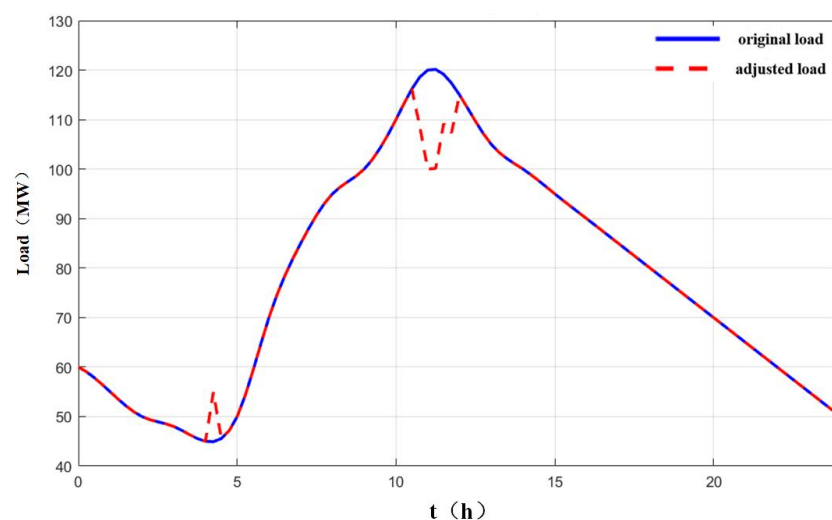


Figure 8. Changes in load curves under variable-power strategy.

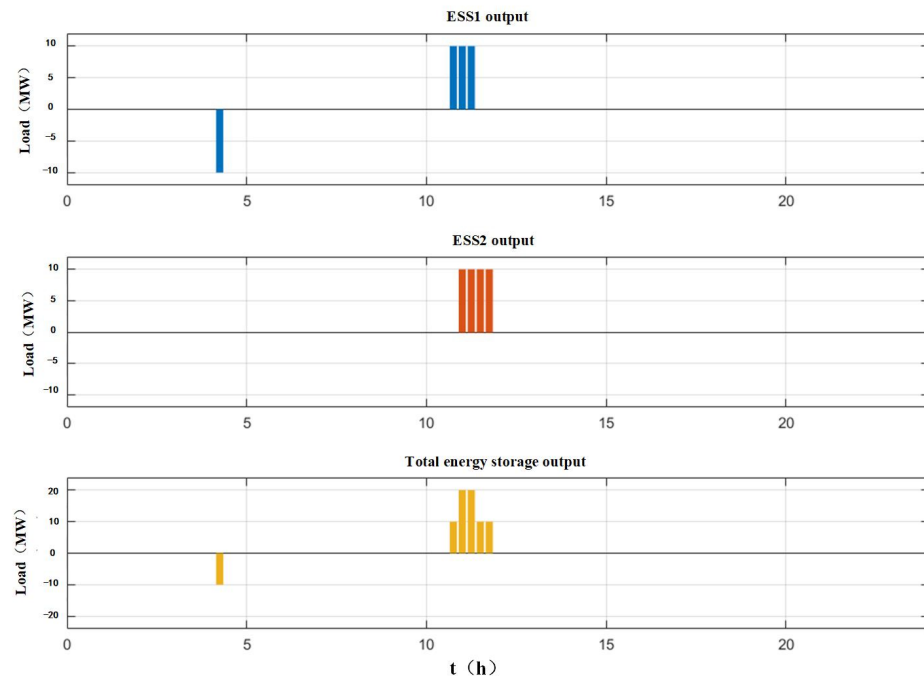


Figure 9. Schematic diagram of BESS output under variable-power strategy.

Table 4. Output status of the BESS.

Output Stage	Charging Period	Discharging Period
Total duration	3.75–4.25 h (0.5 h)	10.75–11.75 h (1.0 h)
Stage 1	Charging two BESS simultaneously (3.75 h)	Discharging only BESS1 (10.75 h)
Stage 2	Charging only BESS2 (4.00 h)	Discharging two BESS simultaneously (11.00 h)
Stage 3	Charging only BESS1 (4.25 h)	Discharging only BESS2 (11.50 h)

Table 5. Summary of energy storage peak-shaving effects under variable-power strategy.

Indicator	Before Peak Shaving	After Peak Shaving	Change
Maximum load (MW)	120.13	114.50	5.63
Minimum load (MW)	44.92	47.20	2.28
ΔP (MW)	75.21	67.30	7.91
α	0.3739	0.4122	0.0383
β	0.6261	0.5878	0.0383
X	22.8483	21.2826	1.5657

Figure 8 shows that after adopting a multi-ESS coordinated variable-power strategy, the load curve is precisely constrained within the target band. Compared with Figure 6, significant improvements can be seen: during the midday off-peak period, three-stage charging continuously raises the valley value to 47.2 MW; during the evening peak period, time-of-use discharge stabilizes the peak value at 114.5 MW. This result demonstrates that equivalent continuous-power regulation can be achieved through a discrete constant-power BESS cluster, significantly reducing the standard deviation of the daily load.

Figure 9 illustrates the operational strategy of multi-BESS coordination: two BESS units alternate operations to achieve equivalent variable power. During the charging phase, the power gradually transitions between -20 MW (dual unit) and -10 MW (single unit); during the discharging phase, the power stops between 10 MW and 20 MW. This dynamic scheduling achieves continuous changes in charging and discharging power through a

combination of start–stop operations, increasing the total regulation duration to 2 hours, which is a 33% improvement over the constant-power strategy, thereby fundamentally enhancing operational efficiency.

Table 4 quantifies the scheduling logic for multi-BESS phased coordination. Taking the discharge period as an example, Phase 1 involves only BESS1, Phase 2 involves dual-machine parallel operation, and Phase 3 switches to BESS2. This sequential combination of discrete devices is the key to achieving the smooth load curve shown in Figure 8, validating the idea of optimizing cluster behavior to precisely track the target band. The duration of each phase is ≤ 0.25 h (minimum scheduling interval), demonstrating the strategy's ability to respond to short-term fluctuations.

Table 5 data indicates that the variable-power strategy has achieved a significant overall improvement in peak-shaving performance, with the peak-to-valley difference reduction increasing to 7.91 MW, reflecting better peak-shaving and valley-filling effects. Additionally, the standard deviation of load decreased by 1.57, representing a 60% improvement. This highlights the advantage of the variable-power strategy in enhancing the depth and precision of peak shaving and valley filling. More importantly, the peak-to-valley coefficient has risen to 0.4122, whereas the peak-to-valley difference rate has decreased to 0.5878. This indicates a higher degree of curve flattening and a smaller range of load fluctuations, thereby better meeting the requirements of the evaluation criteria.

The multi-BESS coordinated variable-power strategy generates equivalent continuous adjustable power at the grid level through the combination of start/stop sequences of discrete constant power units, achieving precise load tracking. As shown in Figure 8, the load curve is strictly constrained within the target band: during the midday off-peak period, three-stage charging continuously raises the valley value from 44.92 MW to 47.2 MW; during the evening peak period, time-of-use discharge precisely suppresses peak fluctuations.

The performance of the constant and variable-power control strategies is inherently dependent on the configured BESS parameters, primarily the rated power and total capacity. It is critical that these core BESS parameters are kept constant in order to provide a fair comparative analysis of the two fundamentally different control strategies under identical conditions. The results in Tables 3 and 5 and Figures 6–9 show that the proposed coordinated variable-power strategy utilizing multiple BESS units achieves significantly superior peaking performance compared to a single BESS constant power strategy, even when using the same underlying BESS resources. This highlights the inherent advantages of the coordination mechanism and dynamic scheduling achieved by the variable-power strategy.

3.3. Scalability Analysis for Multiple BESSs ($N > 2$)

The core innovation of the proposed coordinated variable-power strategy lies in its scalable architecture for aggregating N discrete BESS units ($N > 2$) to achieve grid-level equivalent continuous-power control. While the case study in Section 3.2 employs $N = 2$ for clarity of demonstration, the methodology is inherently extensible to larger clusters. Key advantages of larger clusters include the following:

- (1) **Finer Power Granularity:** The equivalent power resolution improves with N , enabling tighter tracking of load fluctuations.
- (2) **Enhanced Flexibility:** Combinatorial control of N units expands the solution space, extending cumulative regulation duration and reducing individual stress via dynamic power sharing.
- (3) **Algorithmic Tractability:** The phased control algorithm (Section 2.1.2) maintains complexity, ensuring real-time feasibility for practical N values.

While validated for $N = 2$, the strategy's combinatorial control architecture ensures native extensibility to arbitrary N . This establishes a scalable foundation for large-scale power-grid peak-shaving applications.

4. Conclusions

This paper proposes and validates a coordinated variable-power control strategy for multiple battery energy storage stations (BESSs) to address large-scale peak shaving in power grids. The core innovation lies in coordinating discrete BESS units to achieve equivalent continuous adjustable power, significantly enhancing flexibility and utilization. The specific conclusions obtained are as follows.

- (1) This strategy addresses the peak-shaving challenge by coordinating the start-stop sequence of multiple constant power BESS units. This coordination produces equivalent continuously adjustable charging and discharging power at the grid level, enabling fine-grained aggregation and control of discrete storage devices. This paper illustrates the basic design principles of this coordinated variable-power strategy and develops an auxiliary optimization model that incorporates load forecasting, security constraints, and evaluation metrics.
- (2) The effectiveness of the proposed strategy is rigorously demonstrated through a comparative case study. The validation clearly demonstrates the superior overall peak-shaving performance compared to the constant power strategy of a single BESS.
- (3) Quantitative analysis confirms the significant advantages: improved energy storage utilization through phased collaborative scheduling, reduced equipment risk and flexibility constraints inherent in single BESS operation, and reduced system dynamic response requirements through the “cluster-based stand-alone replacement” model. All in all, this provides a reliable solution for large-scale energy storage participation in high-precision peaking applications.

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Abbreviations

The following abbreviations are used in this manuscript:

BESS Battery Energy Storage Station
SOC State of Charge

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